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ORIGINAL RESEARCH/SCHOLARSHIP

# Ethical Impact Assessment: Theoretical Foundations, Methodology, and Applications

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## Abstract

Ethical impact assessment (EIA) is a structured approach to evaluating the ethical implications of technology and innovation. In recent years, it has gained momentum among policy-makers, research funders, and technology developers seeking more systematic ways to address ethical concerns. EIA was developed to fill key gaps in existing ethics and impact assessment frameworks by offering a method capable of identifying both current and anticipated ethical issues, particularly in relation to emerging technologies, and connecting these insights to real-world governance and decision-making processes. EIA is characterized by several defining features: a modular and systematic methodology, a forward-looking orientation, consideration of the entire technology lifecycle, a dedicated focus on emerging technologies, and intentional integration with technology governance. While EIA has seen growing uptake in practice, it remains underexplored in academic literature. This essay introduces EIA to scholarly audiences with the aim of fostering further research and interdisciplinary dialogue. It outlines the conceptual and methodological foundations of EIA, explores its practical application, charts its historical development and diffusion, situates it in relation to other ethics and impact assessment approaches, and reflects critically on its current limitations and future potential.

**Keywords** Ethical impact assessment · Responsible innovation · Technology governance · Ethical methodology · Policy applications

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## Introduction: Ethics Meets Impact Assessment

In the early 2000s, emerging technologies—including those of the digital revolution, genomics and biotechnology, neurotechnology, nanotechnology, and energy systems—confronted policy-makers, industry, and other stakeholders with a growing range of ethical challenges. These included concerns about privacy and surveillance, genetic manipulation, human enhancement, environmental impact, and the fair distribution of benefits and risks. These challenges demanded urgent attention, as technologies were advancing rapidly and posed real-world risks that existing laws and norms were not equipped to handle.

Ethical assessment of these technologies was clearly needed but was marked at the time by a structural gap: although academic work in technology ethics was expanding, it remained largely disconnected from institutional practice. Existing ethics assessment procedures in policy and industry were typically narrow in scope, procedural rather than substantive, and insufficiently forward-looking. There was a need for a more integrated and anticipatory approach that could effectively link ethical reflection with the development, governance, and application of emerging technologies.

The objective of this paper is to examine *ethical impact assessment (EIA)* as a response to this need and to consolidate its conceptual and methodological foundations. The paper aims to provide a systematic account of EIA's origins, theoretical underpinnings, methodology, and applications, and to clarify its relation to existing assessment and ethics frameworks. In doing so, it seeks to establish EIA as a distinctive, structured, and governance-oriented approach to the ethical analysis of technology.

EIA builds on earlier traditions such as *technology assessment* (Grunwald, 2009) and *social impact assessment* (Vanclay, 2003). Additionally, later formulations of EIA were informed by reflexive and participatory insights from *Science and Technology Studies* (Felt et al., 2016) and by orientations associated with *Responsible Research and Innovation* (Stilgoe et al., 2013). Although developed independently, EIA also shares methodological affinities with frameworks such as *Human Rights Impact Assessment* and *Data Protection or Algorithmic Impact Assessments* (Danish Institute for Human Rights, 2020; Reisman et al., 2018), which similarly aim to anticipate and address ethical and societal risks. Within the *ethics of technology* tradition (Hansson, 2017), EIA can be understood as a procedural framework that translates ethical reflection into a systematic, anticipatory, and governance-oriented method (Swierstra & Rip, 2007; Brey, 2012).

In earlier, broader assessment approaches, such as *technology assessment* (TA) and *social impact assessment* (SIA), ethical considerations were typically treated as supplementary to technical or social analysis and addressed only in later stages or through external expert consultation. Related initiatives in applied ethics, such as bioethics committees, ethics review processes, and value-sensitive design, helped bring ethics closer to practice, though most remained domain-specific or focused on particular stages of technological development. EIA builds on these efforts by embedding ethical analysis as a core element of assessment, applicable across technologies and oriented toward the entire innovation lifecycle. It thereby integrates

normative reasoning with participatory and anticipatory practices, linking ethical reflection more directly to decision-making and governance.

EIA is a structured, anticipatory process for identifying and evaluating the potential ethical implications of emerging technologies and planned research or innovation projects, with the goal of informing design, governance, and responsible decision-making. As a form of ethics assessment, it contributes to decision-making in research, development, and policy by systematically analyzing and communicating ethically relevant issues. However, EIA extends beyond traditional ethics assessment methods in several important respects.

First, EIA has a *lifecycle orientation* in that it considers ethical issues across all stages of technological development, from design to long-term use. This is reflected in its *anticipatory focus*, which emphasizes early identification of ethical concerns to enable timely and effective responses. Second, EIA advances *ethics-guided governance*: it is designed to embed ethical reflection directly into governance and decision-making processes, moving beyond advisory or compliance-based models. EIA treats ethics not as an external critique but as an integral part of responsible innovation. Third, EIA adopts a *systematized methodology*, offering structured procedures for identifying, analyzing, and integrating ethical considerations across multiple stages of technological development; unlike principle-based or checklist-style reviews, it supports a more systematic, anticipatory, and context-sensitive form of ethical reflection.

EIA is primarily applied by three key stakeholder groups. Policy-makers, including regulators, apply EIA to inform governance, anticipate ethical risks, and ensure that innovation aligns with societal values. Research and innovation funders use it to evaluate the ethical dimensions of proposed projects and to encourage researchers to integrate ethical reflection throughout the research lifecycle. Technology developers use it to identify ethical issues early, manage risks, and align innovation with stakeholder expectations and public trust.

Over the past two decades, EIA has evolved into a mature and formalized methodology. Its maturation is evidenced by growing institutionalization within international ethics frameworks (United Nations Educational, Scientific and Cultural Organization [UNESCO], 2023), integration into research and innovation policy—particularly in Europe under the banner of *Responsible Research and Innovation*—and efforts to standardize procedures (European Committee for Standardization [CEN], 2017). These developments reflect a broader shift toward more systematic, anticipatory, and ethically informed approaches to governing technological change.

While this emerging role of EIA has unfolded in the context of technology governance, it has received limited recognition in the academic literature, both in *ethics of technology* and in related fields such as *science and technology studies*, *technology assessment*, and *innovation governance*. The aim of this essay is therefore to help remedy this situation by offering a comprehensive account of EIA's rationale, theoretical basis, methodology, and applications.

The structure of the essay is as follows. Section "The Emergence and Evolution of Ethical Impact Assessment" traces the emergence of EIA as a distinct approach and examines its uptake by key stakeholders. Section "Theoretical Foundations" discusses the theoretical foundations of EIA, including its definition, objectives, scope,

and normative basis. Section "Methodology of Ethical Impact Assessment" presents the methodology of EIA, detailing its constituent methods and procedural structure. Section "Practical Implementation and Applications" explores practical applications of EIA and the ways in which its results are implemented by commissioning organizations. Section "Ethical Impact Assessment in Comparative Perspective" offers a comparison between EIA and related assessment approaches, and Section "Future Prospects and Conclusion" considers future prospects for EIA and concludes the paper.

## The Emergence and Evolution of Ethical Impact Assessment

EIA combines two established approaches: ethics assessment and impact assessment. Its emergence can be seen as the integration of ethical analysis and assessment with the structured, anticipatory methodology of impact assessment. To understand how EIA developed, it is useful to first examine the historical trajectories of these two foundational approaches.

Ethics assessment in research and innovation (R&I) is a practical activity that aims to identify, analyze, and communicate ethically relevant aspects to inform decision-making in research, development, and policy. The separate consideration of ethical issues in R&I began to take shape in the aftermath of World War II, driven by the recognition of the need for ethics oversight in biomedical research. Landmark developments included the Nuremberg Code (1947) and the establishment of national bioethics committees from the 1970s onward.

In the 1980s and 1990s, ethics assessment expanded as new scientific fields emerged. It became an essential dimension of risk governance for emerging fields like biotechnology, information technology, and nanotechnology. There was an expansion of ethics councils and advisory structures at both national and international levels. Ethics assessment and oversight were developing from a reactive to a more proactive form. However, ethics assessment at this time was largely external to decision-making structures, often involving ad hoc consultation. In the 2000s, awareness was growing that such approaches were insufficient for adequately mitigating the risks of rapidly advancing technologies. As we will see, EIA was one of the responses in light of this realization.

A second key development was the emergence of impact assessment frameworks in governance, beginning in the 1970s and expanding in the decades that followed. In 1969, the United States formalized *environmental impact assessment* (EIA) in law through the National Environmental Policy Act, requiring federal agencies to evaluate the environmental consequences of large-scale development projects prior to approval. EIA is a structured process for assessing potential environmental impacts to inform decision-making and ensure that such impacts are considered in project planning. This U.S. model inspired a wave of similar legislation worldwide, including a 1985 EU directive that mandated environmental assessments within national planning systems. These legal frameworks helped institutionalize environmental governance and encouraged broader reflection on the non-technical consequences of policy and development actions.

*Social impact assessment* (SIA) emerged in the 1970s alongside environmental assessment as a way to analyze the social consequences of major development projects. It arose from the recognition that technological and infrastructure projects often had disruptive effects on communities, livelihoods, public health, and cultural life. SIAs were typically included as components of environmental assessments and remain less standardized. Their relevance increased through institutions like the World Bank and the International Finance Corporation (IFC), which began requiring social assessments in project evaluations, particularly in the Global South. Interest in SIA has also grown in relation to sustainable development and climate adaptation efforts.

A third major form of impact assessment is *technology assessment* (TA). TA is a structured approach for analyzing the potential social, economic, environmental, and, to a limited extent, ethical impacts of technological innovation. Its main purpose is to inform policy and governance by anticipating unintended consequences, identifying societal concerns, and supporting more socially responsible innovation pathways. It emerged in the United States in the early 1970s, most notably with the establishment of the Office of Technology Assessment (OTA), a congressional body tasked with evaluating the societal implications of new technologies. The American model inspired similar initiatives in Europe, such as the Danish Board of Technology and Germany's TAB, which helped institutionalize TA as a systematic practice of anticipatory policy advice.

Traditional forms of Technology Assessment (TA) focused primarily on mapping risks, costs, benefits, and public concerns, without explicit reference to ethical issues or normative frameworks. By the 1990s, TA had diversified into more reflexive and participatory variants that moved beyond this expert-driven model. The most influential among them was *Constructive Technology Assessment* (CTA), developed in the Netherlands (Rip et al., 1995), which reoriented TA from post-hoc evaluation toward actively shaping technological development through early engagement, stakeholder deliberation, and reflection on underlying values.

Related approaches such as *participatory TA* (Joss & Bellucci, 2002) and *interactive TA* (Grin & van der Graaf, 1997) likewise sought to democratize technology governance and incorporate ethical reflection through dialogue and mutual learning. Later developments, including *ethical technology assessment* (Palm & Hansson, 2006) and Grunwald's conception of TA as a mediating practice between ethics and policy (2011), aimed to make this ethical dimension more explicit and systematic by embedding normative evaluation within assessment practice. Although these perspectives significantly strengthened the ethical orientation of TA, in practice ethics often remained an implicit concern—addressed indirectly through discussions of social acceptability or value alignment—rather than through consistent and methodologically robust ethical analysis.

By the early 2000s, the limitations of both ethics assessment and impact assessment traditions had become increasingly evident. Ethics assessment was valued for its explicit focus on ethical issues, but it was often reactive, fragmented, and poorly integrated into the structures and timelines of R&I, making it ill-suited to address the complex and evolving risks of emerging technologies. In contrast, impact assessment was recognized for its systematic and anticipatory character and its embeddedness in

technology governance, but it tended to overlook ethical considerations or address them only superficially, lacking the normative tools needed to engage with deeper moral questions. This contrast prompted calls for approaches that could combine the ethical orientation of the former with the procedural rigor, foresight, and institutional embeddedness of the latter.

The emergence of EIA can be understood in light of these developments, though it is best seen as part of a broader response to growing awareness of societal challenges in technological innovation. By the early 2000s, it had become evident that research and innovation systems were ill-equipped to anticipate and address ethical issues and insufficiently responsive to societal needs and concerns. In Europe, this recognition led to the development of a new governance approach known as Responsible Research and Innovation (RRI), as discussed in the introduction, aimed at aligning R&I processes with public values, expectations, and societal goals.

RRI holds that science and technology should not only meet standards of scientific excellence and economic competitiveness but also be ethically acceptable, socially desirable, and environmentally sustainable. These societal considerations should be integrated directly into the governance structures of R&I systems. RRI was adopted by the European Commission as a central element of its 2014–2020 framework, Horizon 2020, with the active involvement of key stakeholders, including academic institutions.

RRI is commonly described as encompassing five central dimensions that should guide and shape R&I processes:[1]

(1) *Anticipation* – the systematic exploration of potential impacts, risks, and unintended consequences of R&I before they occur, using methods such as foresight studies, scenario building, and impact assessment.
(2) *Inclusive engagement of stakeholders* – the participation of diverse actors, including policy-makers, industry, funders, scientists, civil society, and the public, through participatory and deliberative processes.
(3) *Reflexity*– the establishment of structures and processes that enable critical self-examination among researchers, innovators, and institutions regarding their assumptions, values, purposes, and potential implications.
(4) *Responsiveness* – the capacity of R&I systems to adjust their direction in response to new knowledge, public concerns, or evolving societal needs.
(5) *Alignment with societal values* – ensuring that science and innovation respect ethical principles, human rights, and broader commitments to social desirability and sustainability.

The governance of research and innovation should operationalize these five dimensions through multi-stakeholder collaboration and collective responsibility for outcomes. Governments play a key role in convening stakeholders, setting shared agendas, and creating institutional mechanisms for implementation.

[1] The first four dimensions, defined by Stilgoe et al. (2013), have since become accepted as core elements of RRI. The fifth, alignment with societal values, is similarly central across many accounts.

Work in RRI has led to numerous approaches and instruments, including governance and policy frameworks, tools for improving R&I processes, indicators and monitoring mechanisms, and capacity-building initiatives. EIA can be understood as one such approach developed within the RRI context—specifically within the category of assessment and evaluation frameworks—distinguished by its explicit focus on ethics assessment.

Having outlined the context for the emergence of EIA, we now turn to the development of EIA itself. The term "ethical impact assessment" was first introduced by British scholar and entrepreneur David Wright in his 2011 article, *A Framework for the Ethical Impact Assessment of Information Technology* (Wright, 2011). In this article, Wright proposed a structured methodology for assessing the ethical implications of information technology projects. The approach is designed to be versatile, applicable to the assessment of a wide range of technologies, services, projects, policies, and programmes. While primarily intended for use by technology developers, it is also suitable for other actors, such as policy-makers and decision-makers in R&I governance. Its objective is to ensure that ethical considerations are systematically addressed prior to the deployment of new technologies, thereby mitigating potential risks and taking stakeholder perspectives into account.

Wright's approach is built on a foundation of ethical principles and subprinciples that include the four ethical principles outlined by Beauchamp and Childress (1979): respect for autonomy, nonmaleficence, beneficence, and justice, supplemented by a fifth principle of privacy and data protection. The approach involves several steps, including the identification of ethical issues, the involvement of stakeholders, the deployment of ethics assessment and stakeholder engagement tools, review and audit, and subsequent decision-making based on the outcome of the assessment. Ethics assessment tools that are proposed include checklists of questions and several participatory tools involving experts and stakeholders, including ethical matrix, ethical Delphi, consensus conferences, and citizen panels.

An EIA approach was also proposed in a 2012 paper titled *A Refined Ethical Impact Assessment Tool and a Case Study of its Application* by American researchers Michael Bailey, Erin Kenneally, and David Dittrich (Bailey et al., 2012). This paper presents a structured framework for ethically assessing IT projects using ethical principles, stakeholder analysis, ethical risk assessment, and accountability structures. Several other anticipatory and governance-directed ethics assessment approaches were also proposed in the early 2010s, including anticipatory technology ethics (ATE) (Brey, 2012) and the techno-ethical scenarios approach (Boenink et al., 2010). Earlier, in 2006, Sven-Ove Hansson and Elin Palm proposed ethical technology assessment (Palm & Hansson, 2006).

Wright's approach to EIA can be seen as an important precursor and conceptual foundation for the more structured and institutionalized approaches that were subsequently developed in the context of RRI and the EU Horizon 2020 framework. It laid the foundation for EIA as a systematic and anticipatory practice. Later versions significantly advanced and formalized the approach, broadening its scope, embedding it in institutional contexts, and aligning it with the principles of responsible R&I.

Two subsequent versions of EIA were developed within the EU Horizon 2020 framework: one in the SATORI project (2014–2017), which focused on ethics assess-

ment, and another in the SIENNA project (2017–2021), which aimed to establish ethical and legal frameworks for emerging technologies, specifically AI, human genomics, and human enhancement. Both were large international collaborations involving more than a dozen partners and budgets of around €4 million. A Scopus survey confirms that no other general EIA frameworks have been proposed beyond those developed in the SATORI and SIENNA projects, underscoring their continuity with Wright's original work.[2] David Wright served as a principal investigator in both initiatives, and an explicit agreement—documented in the project reports—was made to integrate the strengths of Wright's EIA with Brey's anticipatory technology ethics (ATE) approach (Brey, 2012).[3]

The initial version of EIA developed in the SATORI project was described in detail in a project report (Brey et al., 2017), and it was also formalized into a CEN Workshop Agreement (CWA), which is a consensus-based reference document that is capable of functioning as a standard (CEN, 2017). In the SIENNA project, the EIA approach was applied on a large scale through EIAs in the fields of AI and robotics (Jansen et al., 2019), human enhancement (Jensen et al., 2020), and human genomics (Soulier et al., 2019). Insights gained from these applications were used to refine and improve the EIA methodology itself (Brey et al., 2022).

Discussion in this paper will concentrate on the version of EIA that was developed in EU framework funding projects. For convenience, Wright's EIA approach will be designated EIA 1.0, while the SATORI version will be termed EIA 2.0 and the SIENNA version EIA 2.5—not 3.0, because the SIENNA version only contains incremental improvements over the SATORI version. Most of the analysis that follows applies to both the 2.0 and 2.5 versions, but when needed, a distinction will be made.

EIA has also gained prominence as a method for assessing the ethical implications of AI. This began with its full-scale application to the field of AI in the SIENNA project (Jansen et al., 2019). As a direct spin-off of SIENNA, EIA was incorporated into the ethics guidelines for AI used by the European Commission in its ethics review process for Horizon funding programmes—guidelines that were co-developed with members from the SIENNA consortium (European Commission, 2021).

A major subsequent development was the publication of UNESCO's EIA tool for AI in 2023 (UNESCO, 2023), designed to support member states in implementing the ethical principles set out in UNESCO's 2021 Recommendation on the Ethics of Artificial Intelligence (UNESCO, 2021). Incidentally, UNESCO was a participating organization in the SATORI project and as such, had already been involved in the development of its EIA approach, and the head of its ethics unit also served as

[2] A Scopus search conducted on October 25, 2025, for articles containing "ethical impact assessment" or "ethics impact assessment" in the title, abstract, or keywords yielded 43 results. Of these, 27 were authored by or referenced Wright or originated from, or referenced, the SATORI or SIENNA projects. Ten merely mentioned the term without substantive development, two applied it to non-technological domains, and four proposed methods limited to information technology.

[3] The strengths of Wright's approach, less pronounced in Brey's ATE, were its focus on institutional usability, its procedural clarity, its focus on risk mitigation, and its compatibility with other impact assessment approaches. The strengths of Brey's ATE approach were its strong anticipatory scope, normative depth, contextual sensitivity, and attentiveness to the distinct ethical features of different technologies.

an advisor to the SIENNA project. The UNESCO tool is structured around guiding questions and assessment prompts, making it well-suited for policy and administrative use. Compared to the SATORI/SIENNA approach, it places less emphasis on in-depth ethical analysis, context sensitivity, structured stakeholder engagement, and systematic anticipatory methods. The remainder of this paper focuses on the SATORI/SIENNA approach as the basis for analysis.

## Theoretical Foundations

This section outlines the theoretical foundations of EIA, starting with its definition, scope, and primary objectives. This will be followed by an account of its normative-evaluative foundations, which justify the soundness of the ethical judgments it produces, and its normative-procedural foundations, which justify the criteria for the processes through which these ethical assessments are generated.

### Definitions, Scope and Objectives

As stated earlier, EIA can be defined as follows:

> *Ethical impact assessment* is a structured, anticipatory process for identifying and evaluating the potential ethical implications of emerging technologies and planned research or innovation projects, with the goal of informing design, governance, and responsible decision-making.

Let us now further unpack this definition. First, EIA is a form of ethics assessment. *Ethics assessment* is the systematic identification, analysis, and evaluation of ethically relevant aspects of R&I practices, projects, policies, or institutions, with the aim of informing decision-making and ensuring alignment with ethical principles, normative standards, and societal values. Ethics assessment is thus a broader practice, of which EIA represents a specific, anticipatory approach focused on evaluating the potential ethical implications of emerging technologies and planned innovations. Ethics assessment should be distinguished from *ethical analysis*, a more general and typically academic activity focused on examining ethical issues and principles outside of formal procedures or institutional frameworks. Ethical analysis often serves as a methodological foundation for ethics assessment.

*Ethics assessment* is the systematic identification, analysis, and evaluation of ethically relevant aspects of research, innovation, technologies, or policies, aimed at informing decision-making and aligning practices with ethical principles and societal values. It employs analytical and participatory tools such as ethical risk analysis, stakeholder consultation, ethical matrixes, Delphi studies, and scenario methods to identify and evaluate moral issues and impacts. *Ethical guidance*, by contrast, provides structured advice, frameworks, and methodologies—such as ethics guidelines, professional codes of conduct, checklists, and process-oriented approaches like Ethics by Design or Value Sensitive Design—that help actors deliberate on ethical challenges and translate assessment outcomes into concrete recommendations

or operational measures. While the two are complementary—assessment generates ethical insight and guidance applies it—they can also function independently, with some bodies issuing guidance without a formal assessment or conducting assessments without prescribing actions.

While EIA is primarily an ethics assessment approach, it can also serve a guidance function when required. In the EIA 2.0 model, this function was built in through a final "remedial actions" stage, where assessors formulated recommendations to address identified concerns. In the EIA 2.5 model, this stage is optional, reflecting a more flexible, context-sensitive approach that acknowledges some commissioning bodies may prefer to interpret and act on results internally, without the assessor's help. To be clear, EIAs are not conducted for reflection alone but to support responsible decision-making, though the assessor's contribution to that process through concrete recommendations is not a fixed requirement.

Other aspects of the definition pertain to what is assessed, who it is assessed for, and with what objective. Let us first consider in more detail the *objects of assessment*. EIA is intended to be a highly versatile and scalable approach that can accommodate multiple objects of assessment. Objects of assessment include:

(1) *Emerging technology fields and subfields* (e.g., nanotechnology; surgical robotics).
(2) *Types of technological artifacts*, *systems and processes* (e.g., drones, mobile networks, food irradiation procedures).
(3) *Specific applications or use cases of new technologies* (e.g., AI for predictive policing; nano-coatings; digital identity verification).
(4) *Programmes and projects in R&I* aimed at making advancements in any of the previous three areas, including both those that carry out R&I (performing) and those that support or finance it (funding).

With modifications, EIA could also ethically assess technology policies, institutions, and governance structures, since these influence the way emerging technologies impact society. For instance, a national regulatory policy for the deployment of facial recognition technology could assess how the policy shapes the use of these technologies and what the potential or likely ethical risks and impacts of this policy are, such as mass surveillance or disproportionate targeting of minority groups.

EIA is not appropriate for all technologies or R&I projects. It is particularly well-suited to emerging and potentially disruptive technologies that involve significant uncertainty, have wide-ranging societal impacts, affect users or social institutions, and raise substantial ethical concerns. For R&I projects, EIA is appropriate only when such technologies are being developed or deployed. It is generally not needed for projects focused solely on exploratory research, those making only incremental improvements to existing technologies, or projects that form part of a larger programme already subject to an overarching EIA.

Next, we turn to the *commissioning entity*, the organization that initiates and funds the assessment. Usually, this is also the user of the assessment. For the assessment of emerging technology fields and types of technological artifacts, the most likely commissioners are government bodies, regulatory agencies, research funders, interna-

tional organizations, and industry associations. They have a strategic, regulatory, or anticipatory interest in shaping the development, oversight, embedding, and deployment of these technologies, and therefore also in understanding and addressing their ethical risks and societal impacts.

For the ethical assessment of specific applications and use cases of new technologies, the most common commissioners are regulators, standards bodies, technology developers, and public clients or procurers (e.g., municipalities, healthcare providers, security services), since they are accountable for the safe and ethically sound deployment of these technologies. For the ethical assessment of programmes and projects in R&I, the most typical commissioners include research performing organizations (e.g., universities and public institutes), private companies and consortia, public R&I funders, and governmental agencies overseeing research, as they have a stake in ensuring responsible conduct, ethical compliance, and societal relevance.

Another key consideration is the role of the *assessor*, as their expertise and institutional affiliation significantly influence the quality and credibility of the EIA. EIAs are best conducted by a team of professionals with complementary expertise, including skills in ethical analysis, foresight, stakeholder engagement, and participatory methods, as well as knowledge of the relevant technological domain and the social, legal, and institutional context in which the technology operates. Assessors should have a clear mandate to conduct the assessment and be sufficiently independent from vested interests to ensure impartiality and trustworthiness. Assessing bodies may include academic or research institutions with dedicated ethics or technology assessment units, independent ethics committees or advisory boards affiliated with public institutions, and ethics consultants or expert panels commissioned by industry, government agencies, or international organizations. Let us conclude with a more precise account of the *objectives* of EIA. As noted in the earlier definition, the purpose of an EIA is to identify and evaluate the potential ethical implications of emerging technologies and research or innovation projects, with the broader aim of informing design, governance, and responsible decision-making. EIA thus has an immediate objective—to identify and evaluate ethical implications—and the final objectives, which are to support responsible decision-making and help align technological development with societal values. However, these final objectives lie beyond the scope of EIA itself and require follow-up actions by relevant stakeholders. The immediate objective is the one that falls within the EIA's direct function. As will be discussed in Section "Practical Implementation and Applications", stakeholders may also assign additional objectives to EIAs that reflect governance or innovation aims beyond ethical responsibility and value alignment.

### Normative-Evaluative Foundations

*Normative-evaluative foundations* are the normative principles, evaluative standards, and moral assumptions that guide the assessment of ethical issues within an ethics assessment approach. In many ethics assessment approaches, normative-ethical foundations take the form of articulated ethical principles, values, or standards that guide the identification and analysis of ethical issues. EIA, however, does not adopt

a formal ethical framework of this sort, instead relying on a pluralistic and context-sensitive orientation.

EIA does not adopt a single normative framework for three main reasons. First, EIA must apply across diverse technologies and R&I practices, each raising distinct ethical issues and requiring different evaluative standards—for example, human control and accountability in autonomous weapons, animal welfare in livestock biotechnology, or human dignity in gene editing.

Second, moral frameworks often lack the flexibility to capture subtle, context-dependent moral concerns or emergent social values. Most formal ethical theories are *principlist*, emphasizing predefined norms such as autonomy, beneficence, or justice. While these provide structure and analytical clarity, they often fail to account for the contextual and evolving nature of moral concerns. Overreliance on fixed norms risks overlooking morally significant nuances and undervaluing the perspectives of those directly affected. Ethical reflection in EIA must therefore integrate experiential, intuitive, and stakeholder-based judgments that reveal values not encompassed by abstract principles.

Third, ethical values and standards vary across societies, cultures, and even organizations. Values such as individual autonomy, solidarity, or sustainability may be weighted differently depending on cultural, institutional, or sectoral norms. A fixed ethical framework may impose ethical standards that lack legitimacy or relevance in certain contexts, leading to resistance or ethical blind spots. By avoiding commitment to a single formal framework, EIA can adapt to different settings, enabling more inclusive, context-aware assessments that respect plural moral landscapes.

Still, EIAs often rely on formal ethical frameworks, but these are adopted based on context-sensitive criteria and supplemented with informal ethical methods that include stakeholder consultation, expert judgments, value mapping, and others. Regarding the use of formal frameworks, it is recommended that international human rights law, together with relevant national constitutional and regulatory provisions, be recognized as foundational normative sources for ethics assessment. Ethics guidelines that have broad stakeholder support, such as ethics guidelines for AI in genomics research, can also be utilized. In addition, it is recognized in the EIA that human rights, justice, sustainability, accountability, and the avoidance of harm and promotion of well-being are widely accepted core concerns of ethics assessment. These values serve as broadly accepted normative reference points that help structure ethical evaluations across diverse domains, technologies, and cultural contexts. As such, they should be explicitly integrated into the assessment process.

To an extent, then, EIA leaves it to the assessor to assemble a set of formal and informal standards and methods for ethics assessment. EIA contains a set of procedures and recommendations for helping the assessor make choices. The normative standards adopted in an EIA depend on the nature of the technology or R&I practice that is assessed, the national or cultural context in which the EIA is undertaken, and the availability of time and resources for the assessment. Time and resources are relevant because they determine whether certain assessment instruments can be used, such as extensive stakeholder or expert consultation. Invariably, the assessor will use a number of complementary frameworks and methods to identify and evaluate ethical impacts, typically drawing from international human rights law, widely shared

ethical principles and moral intuitions (both general and specific to the technology or practice), stakeholder values, and context-specific societal norms.

### Normative-Procedural Commitments

The normative commitments of an ethics assessment approach are reflected both in its reliance on substantive ethical principles and in its *normative-procedural commitments*: normative standards that govern how the ethics assessment process ought to be conducted. The structure of the EIA process embodies several such commitments, some of which are explicitly stated as desirable features of an EIA, while others are left implicit.

A first normative-procedural principle is *transparency*: the procedures, criteria, and justifications used in the ethical assessment should be clearly communicated and accessible, thus expressing a commitment to accountability, openness, honesty, and trust in the process and its outcomes. A second principle is that of *inclusiveness*: the idea that all relevant stakeholders should have the opportunity to participate in the assessment project, thus supporting moral recognition and respect and democratic legitimacy. Third, *fairness in deliberation* is aspired to, which implies a commitment to impartiality, equal participation, and reasoned dialogue, with no room for dominance by power imbalances or ideological bias.

*Reflexivity* is a fourth principle. It entails critical reflection by those conducting the assessment on their implicit assumptions, value commitments, institutional influences, and potential biases, so as to ensure fairness and ethical rigor. Fifth, *responsiveness* is a guiding principle, meaning the capacity of the assessment process to adapt to new information, stakeholder feedback, and changing social and technological contexts. Together, these principles affirm that ethical assessments in EIA are shaped through procedures that are fair, transparent, democratic, accountable, and critically reflective, and that ensure both the ethical integrity and social legitimacy of the assessment process.

## Methodology of Ethical Impact Assessment

EIA employs a set of integrated methods aimed at achieving its primary objective: to identify and evaluate the potential ethical implications of emerging technologies and R&I projects. In this section, core methods of EIA will be presented and, where appropriate, justified. It will start with a discussion of the required disciplinary methods needed to carry out an EIA and their interdisciplinary integration. This is followed by a presentation of the methodology proper: the procedures for its preparatory phase, the steps taken in an EIA, and the methods that are involved in these steps.

### Required Disciplinary Methods and Interdisciplinary Integration

EIAs investigate ethical implications of emerging technologies, or of projects that are involved in R&I for these technologies. In EIA, these implications are sometimes also called ethical impacts. An *ethical impact* is a consequence arising from the

development, deployment, or use of a technology that can be evaluated as morally positive or negative based on ethical principles, values, or societal norms. Examples include privacy violations resulting from the use of facial recognition technology and reduced well-being caused by wearable health tracking devices that increase anxiety through constant self-monitoring. An EIA addresses both present ethical impacts arising from the current state of a technology and plausible or potential impacts that may emerge from its further development, deployment, and use.

To carry out an EIA, one needs the capacity to identify possible and probable future consequences of a technology—for users, stakeholders, society, and the environment—as well as the ability to determine which of these consequences are ethically significant and how they should be evaluated. These two core capabilities define the methodological foundation of EIA: *foresight analysis*, typically drawn from the field of foresight studies (Georghiou et al., 2009), and *ethical analysis*, grounded in the discipline of ethics, particularly ethics of technology (Hansson, 2017). Foresight analysis enables the identification of possible and plausible future consequences of a technology, including indirect and long-term effects, while ethical analysis provides the tools to assess the moral significance of these consequences. Notably, foresight analysis may also be integrated into broader approaches such as technology assessment and social and ethical impact assessment.

To carry out a proper foresight analysis and ethical analysis, an understanding of the technology and its social, legal, and institutional context is essential. Foresight analysis requires insight into the technology's capabilities, limitations, and possible trajectories, as well as the environments in which it might operate. This understanding is essential for anticipating realistic and context-sensitive future scenarios and potential impacts. Therefore, expertise is needed from the relevant technology domain(s), as well as from disciplines like science and technology studies (STS), sociology, law, political science, and others, to ground assessments of how and why impacts emerge in specific settings.

Stakeholder engagement and participatory methods constitute a final area of essential expertise in the execution of an EIA. These methods involve the systematic involvement of those who are affected by, have an interest in, or are knowledgeable about the technology or innovation being evaluated. Stakeholder engagement in EIA serves to ensure that ethical assessments are socially grounded, inclusive, and responsive to the understandings, concerns, and values of those affected by or knowledgeable about the technology. To achieve this, participatory methods must identify stakeholder-perceived ethical concerns, capture diverse values, support reflection, enable equal deliberation among participants, and validate findings. Stakeholder consultations shape both the foresight analysis and the subsequent ethical analysis.

The interdisciplinary integration of these methods in EIA is guided by the distinct epistemic and normative requirements of the different steps in the EIA process. Each step demands specific types of expertise and methods. If these are multiple methods, then their integration must proceed in a functional and coherent way. Details on how this happens in EIA will be provided later.

### Preparatory Planning Stage

Having outlined the key methods required in EIA, we now turn to the main steps of the assessment process. The first is *preparatory planning*, conducted before the formal launch of the EIA. At this stage, it is determined whether an EIA is warranted. If so, the scope, aims, and expected outcomes are defined, and the necessary resources, institutional arrangements, procedural conditions, and methods are consolidated into a coherent EIA plan.

To decide whether an EIA is needed for a particular technology or R&I project, an *EIA threshold analysis* is performed. This preliminary analysis determines whether the project is likely to raise ethically significant issues. In the SATORI project, a threshold questionnaire was developed for this purpose (Brey et al., 2017; CEN, 2017). It asks about the nature, consequences, and risks of the technology, and its results can guide an ethics committee or expert in recommending whether to proceed.

In subsequent planning, the *appropriate scale* of the EIA is defined. EIAs are scalable, adjusting to the project's complexity and ethical risk. Three levels can be distinguished: small-scale EIAs by a single assessor or small team; mid-range EIAs with broader methods and stakeholder engagement; and full-scale EIAs with extensive resources, interdisciplinary teams, and comprehensive participation. The choice depends on the severity of ethical issues, the degree of institutional control, and practical constraints such as time, budget, and capacity.

### Step-by-Step Assessment Procedure

Following the completion and approval of the EIA plan, the actual EIA can proceed. An EIA is held to require six distinct steps. There is also an optional seventh step, consisting of recommendations to inform subsequent decision-making. Figure 1 outlines the steps of the process and identifies the types of expertise required at each stage. While this figure could suggest a linear process, EIA is best understood as an iterative process. New ethical concerns, stakeholder input, or new technological or contextual developments may arise at any stage, requiring earlier steps to be revisited.

In the process, the first three steps involve a structured scoping of the object of assessment. During these steps, a clear, structured, and context-sensitive understanding of the technology or innovation under assessment is attained. For technology fields, the account should describe and analyze the technology field in general, its subfields, and its basic techniques and approaches, as well as the types of products and applications that are in existence or are envisioned. In these descriptions, the broader social, cultural, legal, and institutional context in which the technology is developed, deployed, and used should also be considered. For R&I projects, the descriptive analysis would focus on the specific technologies, tools, methods, or infrastructures that the project intends to develop or apply.

Foresight analysis is undertaken in Step 4.[4] During this step, foresight methods are applied to identify current and possible future impacts of the technology at large,

[4] During the earlier scoping phase, foresight analysis could also be applied in order to anticipate possible future developments of the technology and possible future products and applications.

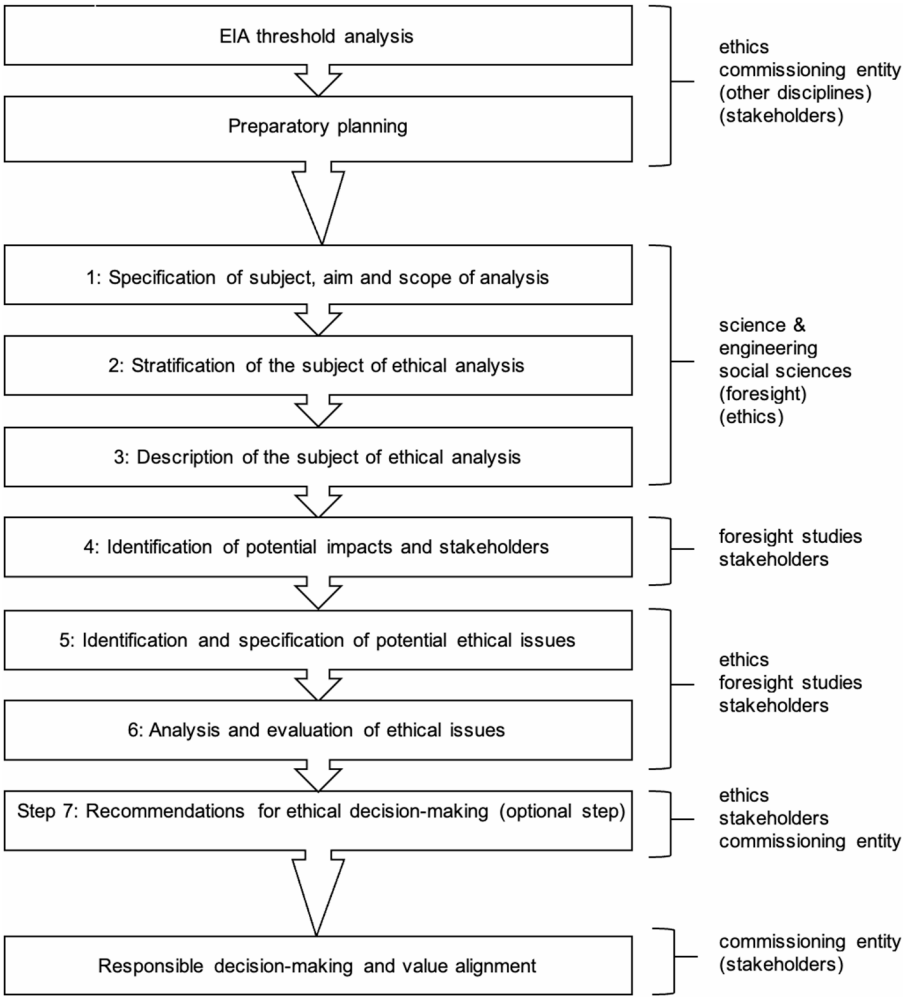



**Fig. 1** EIA steps and required expertise and stakeholders

as well as potential impacts associated with particular products and applications. This foresight study should not randomly identify impacts, but identify impacts in normatively sensitive domains, such as physical health, cognitive and psychological functioning, conditions of employment and work organization, functioning of public institutions, and impacts on ecosystems. In addition, the foresight analysis should consider broader, emerging patterns that may emerge if a technology is adopted at scale and that could be ethically relevant.

While stakeholders would ideally be already involved during the initial scoping stages, a comprehensive identification of relevant stakeholders is only possible at the conclusion of Step 4. This is because those who qualify as stakeholders by virtue of being affected by the technology can only be determined once the full range of expected impacts has been identified.

Ethics comes in during Step 5, in which ethical impacts are identified and described. The identification of ethical impacts can take place in two ways. First, the conventional impacts from Step 4 should be assessed for their propensity to directly or indirectly give rise to one or more ethical issues. For example, if it was found in Step 4 that robots could impact the workplace by taking on co-worker roles, Step 5 would involve an exploration of ethical issues that are likely to result from such a scenario, e.g., harm to well-being due to job displacement, impacts on human autonomy and dignity, issues with fairness in task allocation, and issues with collection of personal data by robots.

A second way of identifying ethical impacts is through a top-down approach, in which the ethicist begins with relevant moral values or principles and, in collaboration with the foresight expert, develops impact scenarios that illustrate how these could be affected. This method complements the bottom-up, consequence-focused approach and may reveal additional ethical concerns that would otherwise be overlooked.

In Step 6, a comprehension-based ethical analysis and evaluation is conducted. The aim of this assessment is to develop a deeper understanding of the moral significance of the ethical issues identified in Step 5. This is achieved through further clarification and contextualization of these issues and by relating them to relevant moral principles, values, norms, or obligations. The analysis also includes the identification of potential tensions or conflicts between values, but without attempting to resolve them at this stage; the focus remains on ethical understanding rather than on normative resolution. Completion of Step 6 concludes the EIA as such. Subsequent steps may include documenting the results in a written report and communicating findings to relevant stakeholders or decision-makers.

The EIA could be extended with an optional seventh step, where the assessor provides recommendations, guidelines, or an action plan to help mitigate harmful ethical impacts—possibly in collaboration with the commissioner and stakeholders. The assessor might also support or oversee the implementation of these measures. If this step is not included, any follow-up on the assessment will be left to the discretion of the commissioner.

### Approach to Foresight Analysis

In the assessment of technologies, the analysis of present and anticipated impacts includes both an evaluation of current effects and a foresight analysis of possible future impacts. For projects, the emphasis is on plausible near- to medium-term futures, particularly those involving impacts of the R&I activities, and especially those associated with the new technology or technological knowledge being developed. The study of current impacts for technologies draws on empirical data from various sources, including existing impact studies, user surveys and studies, usage and performance data, and expert consultation.

For the foresight analysis, a time horizon needs to be chosen. A long time horizon, say more than 20 years, is likely to lead to unreliable results. Customary is a time horizon of 5, 10, or at most 20 years. There are many methods for foresight analysis. There is no commitment in the SATORI/SIENNA EIA to particular methods. Instead,

an inventory of methods is presented along with some of their strengths and weaknesses (Brey et al., 2017, 2022).

*Scenario-based methods* constitute a particularly suitable approach for the assessment of future impacts. Using these methods, structured, plausible narratives of the future are developed to explore how emerging technologies might lead to different outcomes under varying conditions. In this way, potential impacts can be uncovered. Also highly useful are *expert- and stakeholder-based methods* in which collective judgments about the future are arrived at using a structured or semi-structured procedure. Forms include expert and stakeholder panels, workshops and focus groups, multi-stakeholder dialogues, and the Delphi method—a structured, multi-round expert consultation.

*Exploratory and creative methods*, such as brainstorming and visioning, use imaginative thinking and speculative exploration to reveal unexpected developments and broaden the range of possible futures. Other methods that can be used include *trend- and evidence-based methods*, such as trend analysis, horizon scanning, modelling and simulation, and roadmapping. Horizon scanning, for example, involves the monitoring of diverse sources (e.g., academic publications, media reports, patents) to recognize patterns and signal emerging applications, usage patterns, or impacts of a technology.

### Approach to Ethical Analysis and Evaluation

Methods of ethics are used in Step 5 for the identification of ethical impacts and in Step 6 for their analysis and evaluation. To guide these processes, an overall ethical approach is needed that defines both ethical standards of evaluation and ethical methods. Both need to be established prior to Steps 5 and 6. The approach to defining ethical standards of evaluation was already discussed in Section "Normative-Evaluative Foundations" on normative-evaluative foundations. It was found there that these standards are typically composed of international human rights law, widely shared ethical principles and moral intuitions (both general and specific to the technology or practice), stakeholder values, and context-specific societal norms.

By the time Step 5 begins, most of these standards should be in place, so that they can be used for recognizing relevant ethical issues. As explained in Section "Step-by-Step Assessment Procedure", identification of ethical impacts proceeds through two methods. Using these ethical standards, ethical impacts are identified either by examining how conventional impacts raise ethical issues (bottom-up) or by exploring how moral values might be affected in future scenarios (top-down). In this process, ethicists identify ethical impacts using the established standards, while stakeholders offer practical insights and highlight context-specific values and concerns according to their own ethical standards.

In Step 6, the ethical standards established earlier in the project are applied to analyze and evaluate the issues identified in Step 5, using standard methods of applied ethics and further calibrated through stakeholder input.

### Approach to Stakeholder Engagement and Expert Consultation

The participatory approach in the SATORI/SIENNA EIA involves stakeholder engagement across key stages of the assessment, with the aims of ensuring contextual relevance, reflecting a plurality of values, and enhancing democratic legitimacy. Participation occurs primarily during impact identification (Step 4), ethical issue identification (Step 5), and ethical analysis and evaluation (Step 6). Methods include interviews, focus groups, multi-stakeholder panels, and deliberative workshops. They are used to collect experiential knowledge about how technologies are used and experienced, identify ethical concerns, contextualize values used in analysis, and introduce new values and normative standards (Brey et al., 2022). In this way, stakeholder input strengthens both the impact analysis and the ethical analysis. In addition, expert consultation is integrated throughout the process, with experts contributing technical insight and contextual understanding throughout all stages.

## Practical Implementation and Applications

This section focuses on how EIAs are used in practice and how their findings can be implemented by commissioning organizations. In the introduction, three key users of EIAs were identified: policy-makers, R&I funders, and technology developers, defined as actors who engage in applied R&I. It was also defined how EIAs are intended to be applied by them. For policy-makers, EIAs serve to inform and support ethical governance and regulation by identifying ethical concerns and societal risks associated with emerging technologies. For R&I funders, EIAs can guide the design, evaluation, and monitoring of programmes and projects, helping to ensure alignment with societal values, ethical standards, and the principles of responsible R&I. For technology developers, EIAs offer a means to anticipate ethical issues in the design and deployment of new technologies, enabling early mitigation of risks and fostering more socially responsible innovation.

While the ultimate aim of EIAs is to support responsible decision-making and value alignment, key stakeholders may have additional aims as well. EIAs can also be used, amongst others, to anticipate public responses to new technologies, manage reputational and legal risks, foster stakeholder engagement, and contribute to organizational learning and reflexivity. Thus, EIAs can serve multiple purposes for those who commission them.

Let us now consider what is involved in implementing an EIA. Implementation requires the integration of the findings of the EIA into the decision-making processes of the commissioning entity. This involves interpretation of the ethical assessment in light of their goals and responsibilities, taking into account the identified risks, stakeholder concerns, and value conflicts, and then determining whether and how to act on the findings.

For policy-makers, an EIA is typically commissioned and utilized before technology is widely adopted, and its main expected utility is found in guidance for regulatory frameworks, policy design, or public communication. For research funders, an EIA is typically applied in preparation of new funding programmes and to con-

duct ethics review of projects with a broader, more anticipatory approach. Research funders can also require EIA as part of ethics work packages in grant proposals.

For technology developers, it is recommended that an EIA be conducted either prior to or at the outset of an R&D project or process, with the option of updating it as the work progresses. The results can be used to identify ethical risks and to develop recommendations, guidelines, and standards to guide the development process. As Brey and Dainow (2024) suggest, the EIA can also serve as a foundation for an ethics by design approach, in which moral values and ethical concerns are systematically integrated into the design and development process of a new technology.

## Ethical Impact Assessment in Comparative Perspective

A comparison of EIA with other assessment and ethics approaches highlights its distinctive scope and integrative function. Comparisons can be organized along three dimensions: their normative scope (ethical, legal, or social), methodological structure (procedural, principle-based, or empirical), and institutional role (assessment, design, or governance). EIA differs from earlier assessment traditions, technology assessment (TA), social impact assessment (SIA), and environmental impact assessment (EIA), by introducing systematic ethical reflection and normative evaluation into what were primarily empirical and descriptive procedures.

An additional comparison can be made with human rights impact assessment (HRIA) approaches, which emerged in the early 2000s. The UN has published a human rights impact assessment tool (International Business Leaders Forum et al., 2010); the Danish Institute for Human Rights (2020) has issued a widely used HRIA toolbox; and the European Union (2024) has mandated a Fundamental Rights Impact Assessment (FRIA) for the deployment of high-risk AI systems. HRIAs and related tools such as data protection impact assessments (DPIAs) are anticipatory and share EIA's preventive rationale, but they are primarily legal instruments designed to ensure compliance with codified human or data protection rights. EIA, by contrast, integrates these compliance-oriented analyses within a broader ethical framework that considers non-codified values such as fairness, sustainability, autonomy, and social trust. If so desired, it can incorporate HRIA or DPIA modules while extending the analysis to a wider range of ethical and societal concerns beyond rights-based obligations.

In comparison with other ethics assessment instruments, EIA distinguishes itself from traditional approaches such as ethics board or committee assessments, principle-based or checklist-style reviews, legal–ethical compliance procedures, and philosophical–ethical analyses. It stands out for its dedicated focus on emerging technologies, forward-looking orientation, attention to the entire technology lifecycle, structured and systematic methodology, and ambition to embed ethical reflection within broader frameworks of technology governance.

Other approaches share some of these characteristics but remain less comprehensive or less formalized. Ethical Technology Assessment (Palm & Hansson, 2006), Ethical Risk Assessment (Hansson, 2018), and societal or ethical–legal–social readiness approaches (Bernstein et al., 2022; Francis et al., 2023) are important explor-

**Table 1** Comparison of EIA and major related assessment and ethics approaches

| Approach | Ethical focus | Anticipatory / lifecycle | Methodological structure | Governance orientation | Legal orientation |
|---|---|---|---|---|---|
| Ethical Impact Assessment (EIA) | ++ | ++ | ++ | ++ | + |
| Technology & Social Impact Assessment (TA / SIA) | 0 | + | ++ | ++ | 0 |
| Human Rights & Data Protection Impact Assessment (HRIA / DPIA) | + | + | ++ | ++ | ++ |
| Constructive Technology Assessment (CTA) | + | ++ | + | + | 0 |
| Design-embedded ethics (VSD / EbD) | ++ | + | ++ | 0 | 0 |
| Responsible Research & Innovation (RRI) | + | ++ | + | ++ | + |
| ELSA / ELSI programmes | ++ | + | 0 | 0 | + |

Legend: "++" = strong/explicit "+" = moderate "0" = limited or indirect

atory efforts but lack fully developed or widely adopted methodological frameworks. Ethics by Design (EbD) and Value Sensitive Design (VSD) (Brey & Dainow, 2024; Friedman et al., 2006) provide mature and anticipatory methodologies but operate primarily within the domain of design and development rather than governance. They aim to integrate ethical values into design practices rather than to perform systematic assessments. Indeed, Brey and Dainow (2024) explicitly recommend that EbD be informed by an EIA early in the design process.

EIA also relates closely to traditions of research ethics and to the ELSA/ELSI (Ethical, Legal, and Social Aspects/Implications) programmes that preceded it. Like research ethics, EIA aims to ensure that research and innovation are conducted responsibly, but it moves beyond the procedural focus on participant protection and compliance that characterizes ethics review systems. EIA extends ethical reflection to the broader societal and technological implications of research outcomes and the governance of innovation processes. Similarly, ELSA and ELSI programmes sought to identify and analyse the ethical and social implications of emerging fields such as genomics and nanotechnology, yet they typically remained diagnostic and lacked standardized procedures for systematic assessment. EIA can be viewed as a methodological continuation of these initiatives, translating their analytical insights into a structured, anticipatory, and actionable framework that links ethical reflection directly to research and innovation governance.

Responsible Research and Innovation (RRI), finally, functions as a governance framework that seeks to align research and innovation with moral and societal values. While RRI defines a high-level orientation for responsible governance, it must be supplemented by concrete ethics assessment tools. EIA offers such an instrument by operationalizing RRI's ethical dimension and translating its principles of anticipation, inclusion, reflexivity, and responsiveness into actionable assessment procedures.

Taken together, these comparisons show that EIA's distinctive contribution lies in integrating existing approaches into a coherent framework that connects the empirical focus of impact assessments with the normative ambitions of applied ethics and the structural perspective of governance. See Table 1 for a summary of the comparative analysis.

EIA nonetheless faces limitations arising from the scope and nature of its methodology. Its anticipatory orientation depends on the quality of foresight and scenario analysis, which can be uncertain or speculative when dealing with rapidly evolving technologies. The broad ethical framing that gives EIA its integrative strength can also make it challenging to determine the appropriate depth and specificity of normative analysis, sometimes leading to overlap with existing assessments. Moreover, because EIA seeks to balance inclusiveness, transparency, and analytical rigor, it may be more time- and resource-intensive than narrower assessment tools. Finally, its systematic approach requires interdisciplinary and institutional coordination that is not always in place, which can limit consistent application across contexts. These limitations, however, concern implementation rather than the soundness of EIA's conceptual foundations.

Despite these challenges, EIA adds value through its integration of anticipatory ethics, lifecycle orientation, modular structure, and governance readiness. These features enable it to complement and, where appropriate, extend existing approaches by linking empirical analysis and normative evaluation within a single framework. EIA complements technology assessment (TA) and social impact assessment (SIA) by adding systematic ethical analysis and normative evaluation to their largely empirical orientation; human rights and data protection impact assessments (HRIA, DPIA) by broadening their legal and rights-based focus; and ethical approaches such as research ethics, principlist ethics, academic applied ethics, and value-sensitive design by offering an overarching, lifecycle-oriented, and governance-directed process that connects anticipatory ethical reflection directly to decision-making and technology governance.

## Future Prospects and Conclusion

EIA is an ethics assessment tool developed to support the ethical governance of emerging technologies, with key applications in policy-making, R&I funding, and technology development. It is the only impact assessment approach with a focus on ethics, covering the full breadth of ethical risks and impacts that can result from the development, deployment, and use of new technology, and arguably the only ethics assessment approach that combines anticipatory analysis, lifecycle orientation, a structured methodology, and integration with technology governance to systematically evaluate the ethical implications of emerging technologies.

The next challenge for EIA is to achieve broader institutional uptake—beyond early adopters such as UNESCO and the EU Horizon programmes—by embedding it in funding, regulation, procurement, and technology development. This requires stronger institutionalization, standardized procedures, targeted training, clear mandates, and supportive policy infrastructures to ensure consistent quality.

Methodological improvements remain possible in several areas, including the refinement of foresight techniques, the development of more robust methods for addressing value pluralism and resolving moral conflicts—particularly in diverse stakeholder settings—and enhancing the usability of the approach for technology developers and practitioners. Of particular importance is the development of dedi-

cated, standalone versions tailored to different target groups, including policy-makers, research and innovation funding bodies, and technology developers. While the current methodology is versatile, specialized versions will increase its usability and institutional relevance for these stakeholders.

The continued development and institutional embedding of EIA will depend on its capacity to demonstrate practical relevance, maintain methodological robustness, and foster alignment among diverse stakeholder groups. By offering a flexible yet principled and governance-ready approach that responds to the specific needs of policy-makers, research funders, and technology developers, EIA is well positioned to evolve into a key instrument for ensuring that emerging technologies are ethically aligned with societal values from the earliest stages of their development.

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**Disclosure** The author served as coordinator of the SATORI and SIENNA projects mentioned in the paper and was also work package leader and researcher in the work packages responsible for developing the ethical impact assessment approach.

**Competing Interests** There are no competing Interests.

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